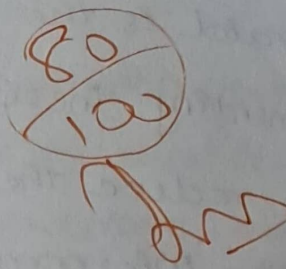


Name:- ccTyoshna

Reg.no:- 192411098

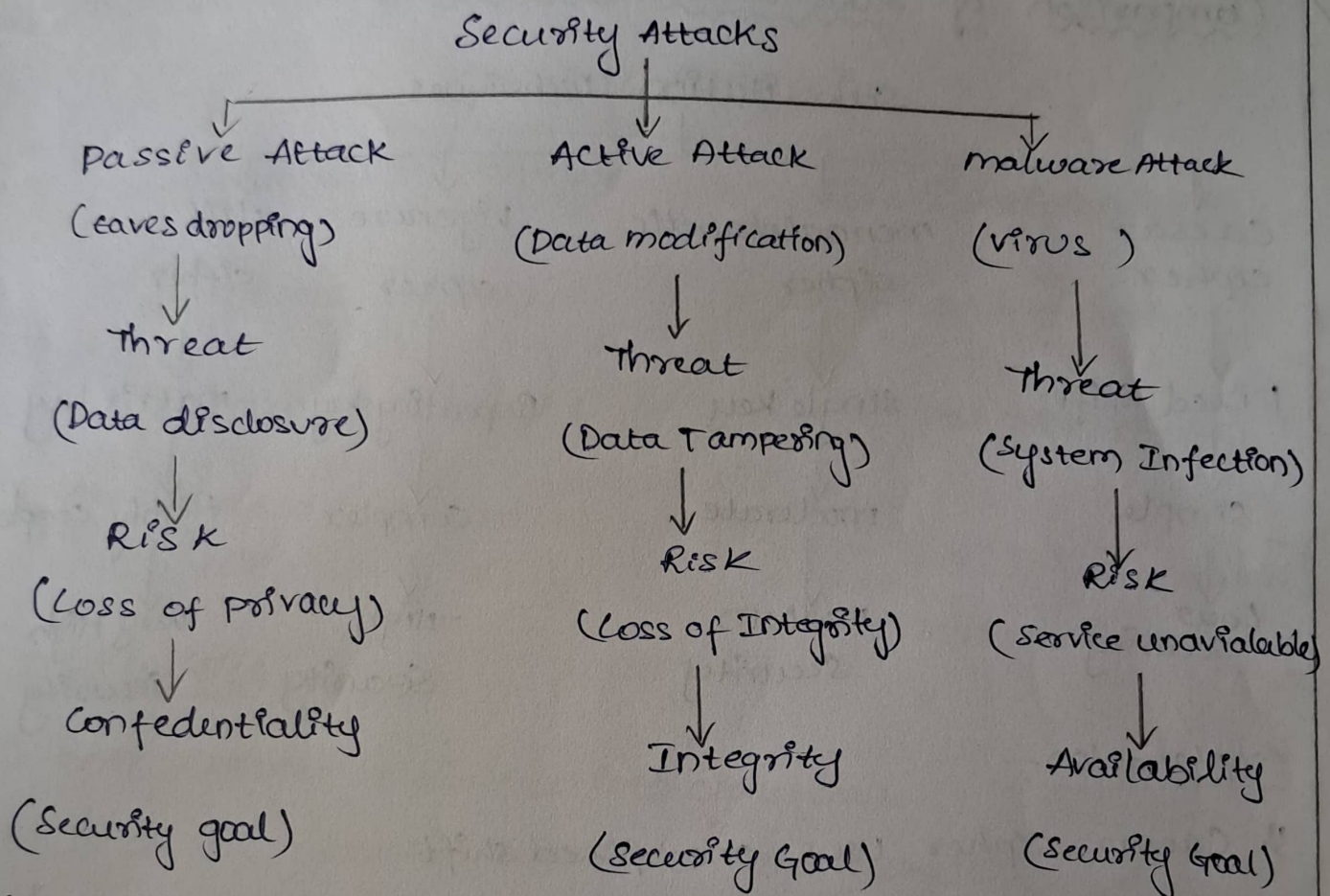
Course code:- CSA5101

Course:- cryptography



CO1: Assessment Tool-2

1) Security Attacks, Threats, Risks and Security Goals

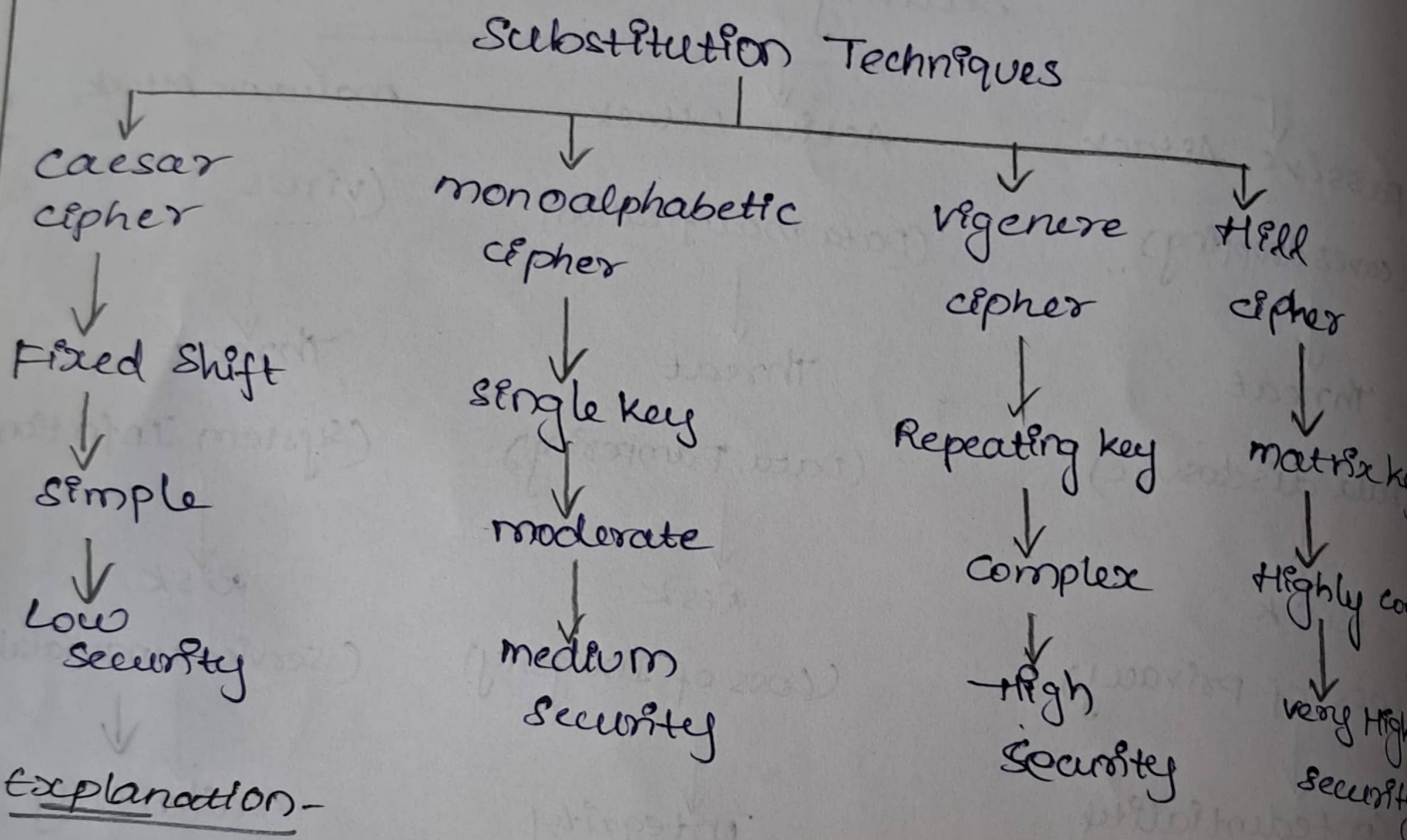


Explanation

- 1) Security attacks attempts to compromise computer systems.
- 2) Attacks create threats that may exploit vulnerability.
- 3) Threats increase the risk of data loss / system damage
- 4) Confidentiality prevents unauthorized access.
- 5) Integrity prevents unauthorized modification.
- 6) Availability ensures systems remain accessible

The CIA triad is the foundation of information security. By implementing these security goals, organizations can reduce the impact of cyber attacks and protect sensitive information.

2) Comparison of Substitution Techniques



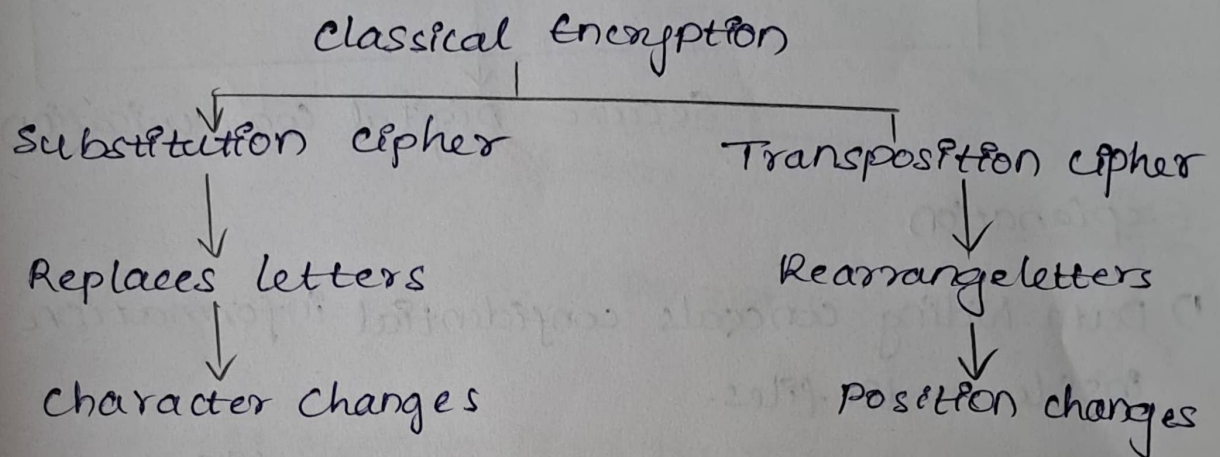
Explanation-

- 1) Caesar cipher uses one fixed shift
- 2) monoalphabetic cipher replaces letters using one substitution table.
- 3) Vigenere cipher uses a repeating keyword.
- 4) Hill cipher encrypts blocks of letters using matrix multiplication.
- 5) Security increases as encryption complexity increases

Analysis -

Among classical substitution techniques, the hill cipher is the most secure because it encrypts groups of letters using matrix operations, making frequency analysis much more difficult

Classical Encryption schemes:-



Ex:- Caesar, Vigenere, Hill

Ex:- Rail Fence, columnar Transposition.

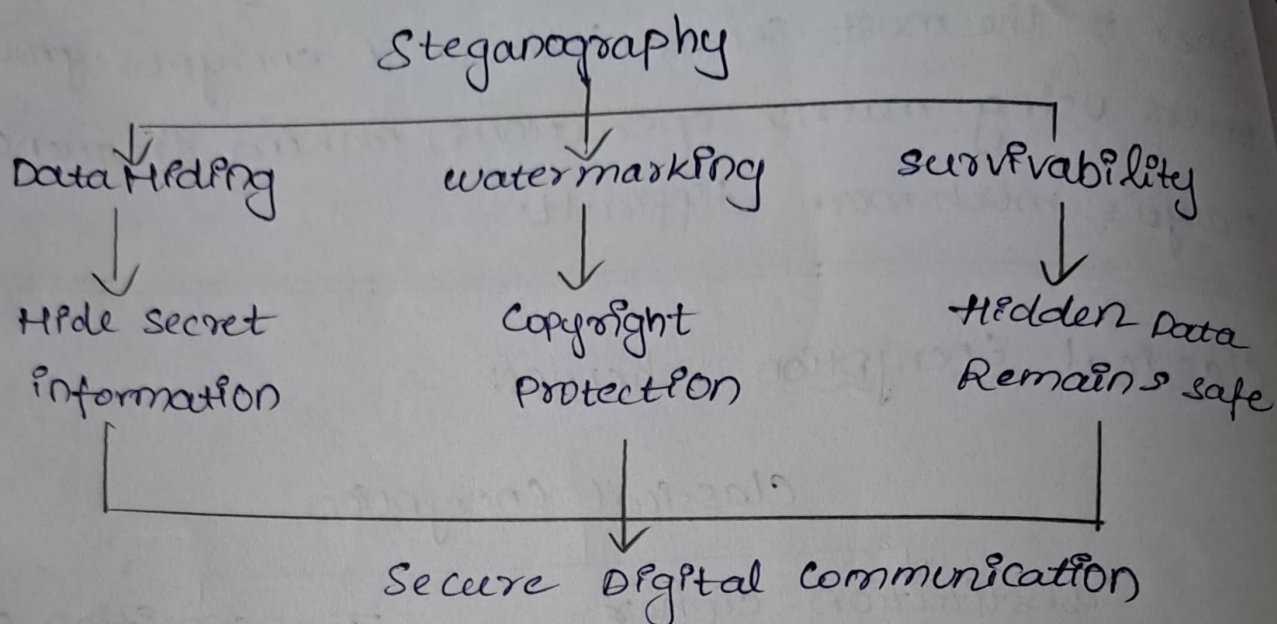
Similarities

- 1) Both convert Plaintext into ciphertext
- 2) Both are classical encryption and decryption.
- 3) Both require encryption and decryption.

Analysis

Transposition ciphers are more vulnerable because the letters remain unchanged. Attackers can use frequency analysis and brute-force methods to recover the plaintext.

4) Steganography



Explanation

- 1) Data hiding conceals confidential information inside media files.
- 2) watermarking protects ownership of digital content
- 3) survivability ensures hidden information remains after compression/editing.
- 4) Together they improve Secure Communication.

Analysis

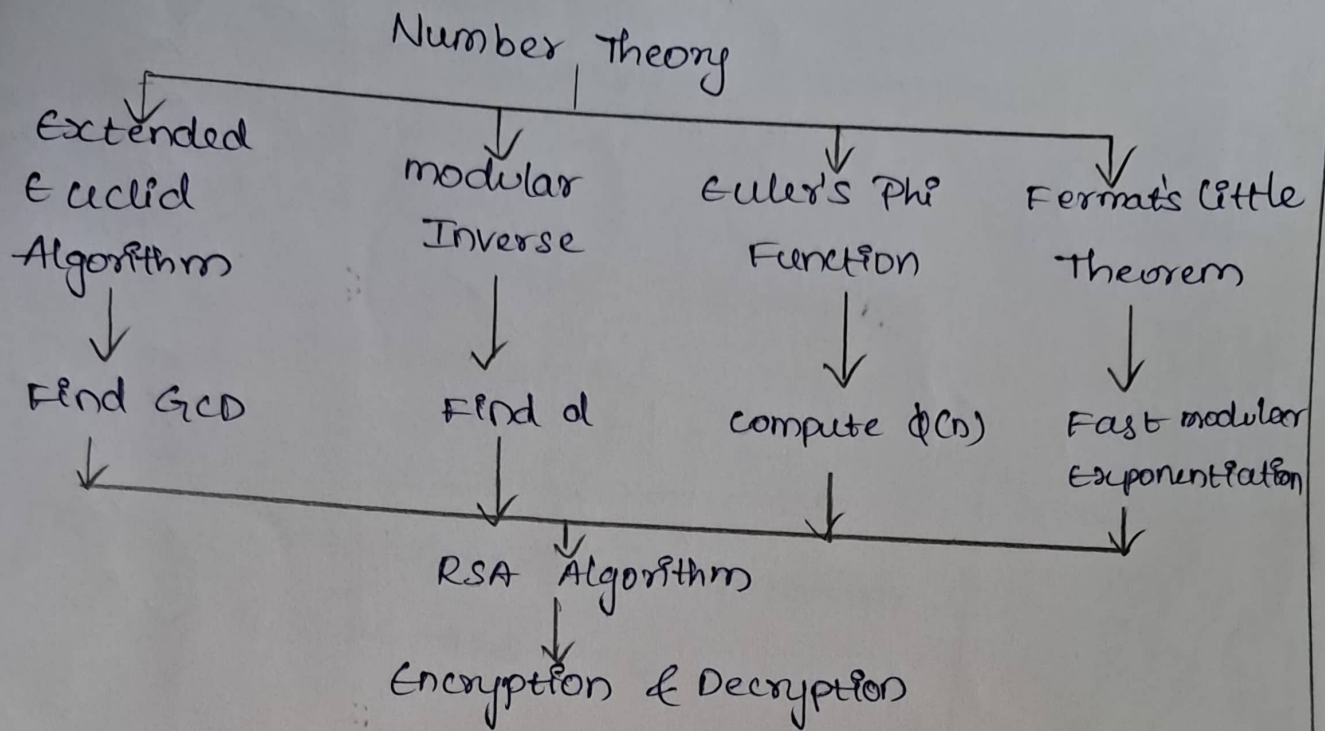
Steganography hides the existence of secret information, making communication more secure than simply encrypting data.

examples

Data Hiding → military communication.

survivability → digital forensic investigation.

Theory in cryptography



Explanation

- 1) Extended Euclid Algorithm finds the gcd & modular inverse
- 2) modular inverse is used to generate the RSA private key.
- 3) Euler's Phi Function is required for RSA key generation.
- 4) Together these concepts form the mathematical foundation of RSA

Analysis

The modular inverse is the most critical concept because RSA decryption depends on the private key generated using the modular inverse. Without it, decryption is impossible.